

not only boost EV performance but also ensure safety and reliability, which are essential for the broader acceptance and success of electric mobility.

A key area in electric vehicle design

A critical aspect of EV development is the electrical and electronics architecture, where optimising harnesses is essential. Efficiently analysing and optimising the harness is essential, not only for EV functionality but also for ensuring the seamless flow of data between vendors and suppliers. This requires robust traceability options to be available throughout the process.

Harness optimisation is particularly complex due to the trend of reducing vehicle weight. The harness system plays a crucial role in this, contributing significantly to the vehicle's weight. Before manufacturing, it is important to analyse and determine the optimum weight of the vehicle, considering the weight contribution of the harness.

Addressing these challenges requires more than just engineering expertise; it demands the support of specialised software that can provide precise inputs for this purpose. One such solution in the market is a renowned harness design and analysis tool. Employing generative and connected engineering approaches, this tool seamlessly integrates with various mechanical computer-aided design (MCAD) systems.

This integration is particularly significant when working with electrical controller area network (CAN) systems. After schematics are prepared, the next step involves routing, which needs to be meticulously planned and executed. The integration of this tool with the design process facilitates efficient routing, making it a vital component in the design and manufacturing of EVs.



An engineering research agency performing aerodynamic testing of a modern eco-friendly electric sports car in a wind tunnel

This solution is widely adopted across the industry, as evidenced by its use in several leading automotive companies. These companies utilise this tool not only for product design but also for optimising manufacturing processes and enhancing performance. The tool's versatility and effectiveness in addressing the unique challenges of electrical and electronics architecture in EVs make it an invaluable asset in the quest to develop more efficient and sustainable EVs.

Aerodynamics and EV design optimisation

Aerodynamics plays a pivotal role in optimising EV design, particularly in minimising drag and improving efficiency. This process involves assessing the vehicle's electronic design and its integration into the overall architecture, ensuring both functional safety and security. As the automotive industry shifts toward modular assembly processes and an increased reliance on purchased components, especially in autonomous vehicles, the need for a collaborative platform to manage these aspects grows increasingly important.

In aerodynamics and structural

design, reducing vehicle weight is a key focus. Additive manufacturing is becoming a significant part of this process, moving beyond traditional manufacturing methods. This new approach not only involves product design and tooling but also extends to optimising aspects like vehicle lighting and the efficient use of space.

One particularly complex product design area is the design of printed circuit boards (PCBs). With numerous components on a single PCB, it is crucial to ensure that the PCB design meets all regulatory norms and is seamlessly integrated into the current design flow. With the development of advanced capabilities in simulation tools, even entry-level engineers can conduct simulations and optimise designs.

A particularly critical aspect of this process is thermal management in PCB design. Sophisticated simulation tools are used to analyse the thermal dynamics within a vehicle. By importing PCB designs into these tools, engineers can simulate and adjust the design, including changing the location of parts to meet thermal management requirements. This process significantly influences the choice of thermal management